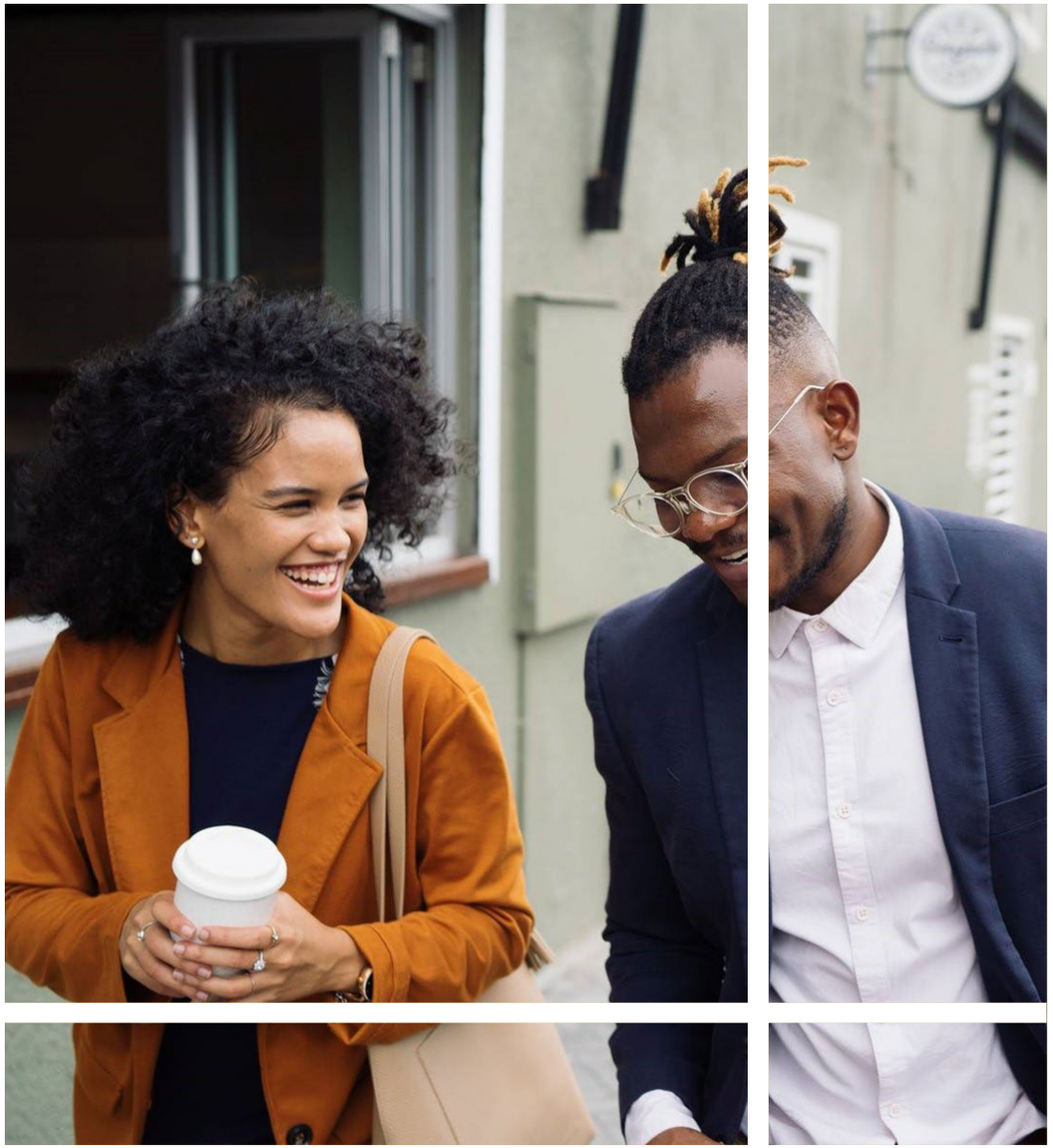
CARESPEAK: A CONVERSATIONAL TRAINING TOOL FOR APRN STUDENTS

NURSING-AI SHOWCASE: CHALLENGE #17 – SIMULATED PATIENT-CENTERED CONVERSATIONS

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AI 395T | **AI in Healthcare** | Prof. Ying Ding, Ph.D.



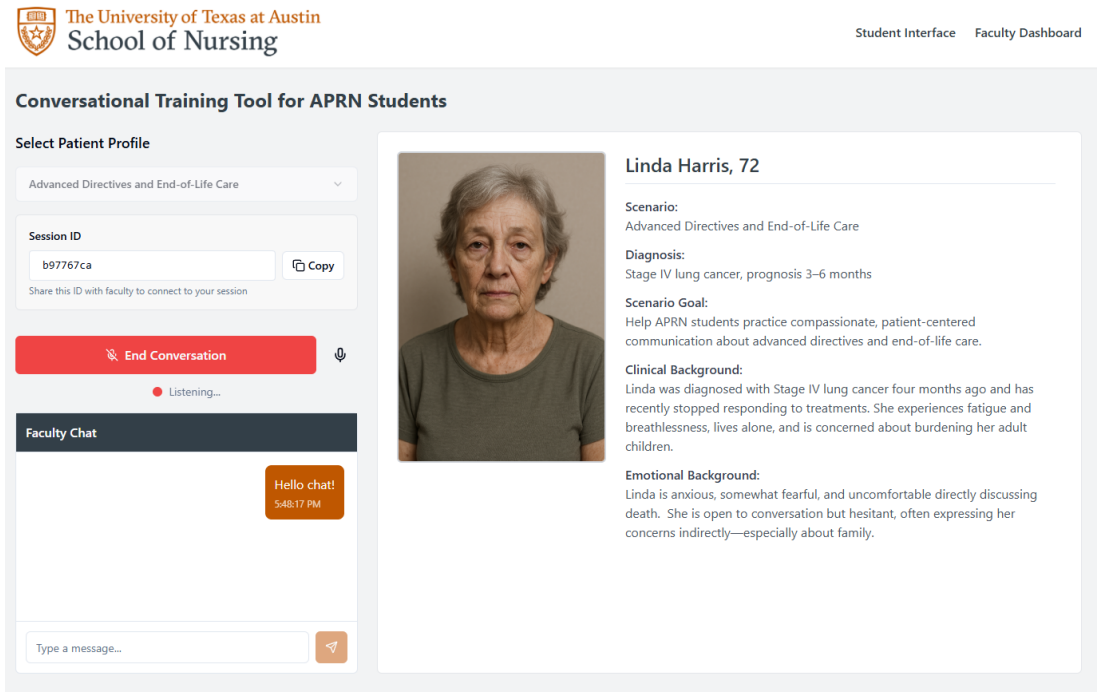


THE NEED: PRACTICING SENSITIVE PATIENT CONVERSATIONS

- **Project:** CareSpeak (Proof-of-Concept for NURSING-AI Challenge #17)
- **Goal:** Help APRN students practice emotionally sensitive conversations (e.g., advance directives, end-of-life).
- **Problem:** Critical skill, but insufficient structured practice; traditional methods (standardized patients) costly & hard to scale.

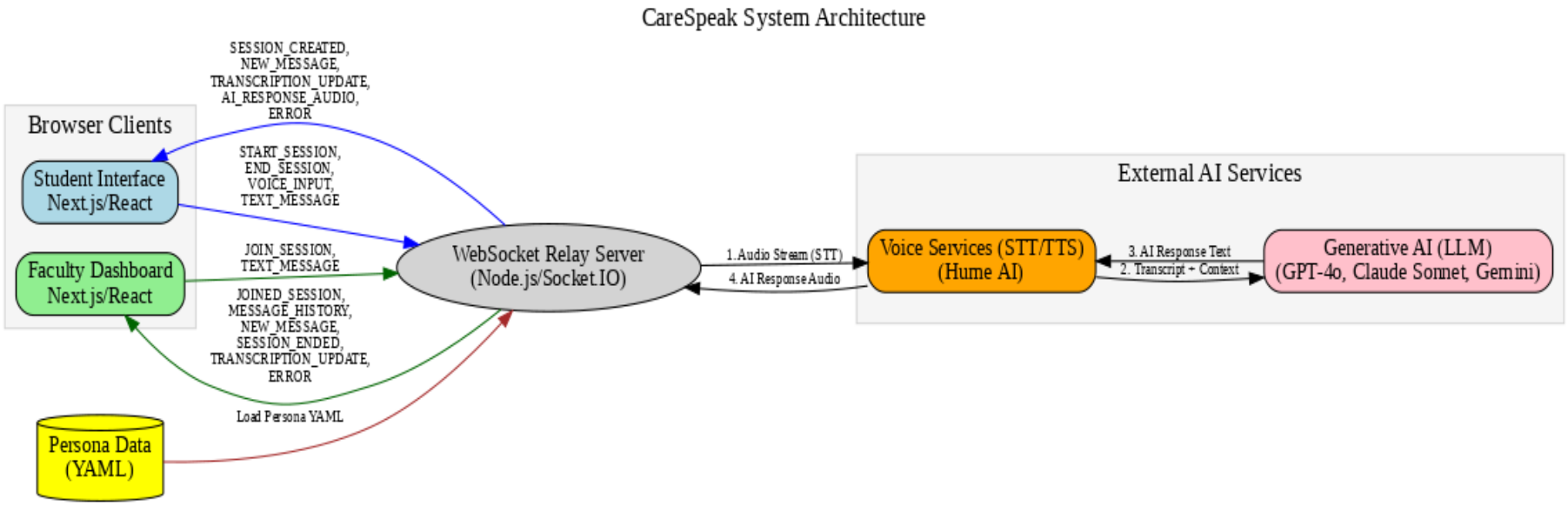
OUR APPROACH:
AI-DRIVEN VOICE SIMULATION

- **CareSpeak:** Real-time, voice-based simulations using natural speech.
- **Affective Computing Focus:** Models how things are said (tone, emotion), not just what.
- Uses Hume AI's affective voice technology for rapid prototype development.
- Addresses skills difficult to teach conventionally.



HOW IT WORKS: REAL-TIME INTERACTION

- Connects Student, AI Patient & Faculty via WebSockets.
  - **Student:** Speaks naturally via microphone.
  - **AI Patient:** Responds with emotionally expressive synthesized speech (LLM-driven).
  - **Faculty:** Observes live (audio + transcript), provides feedback via dashboard.



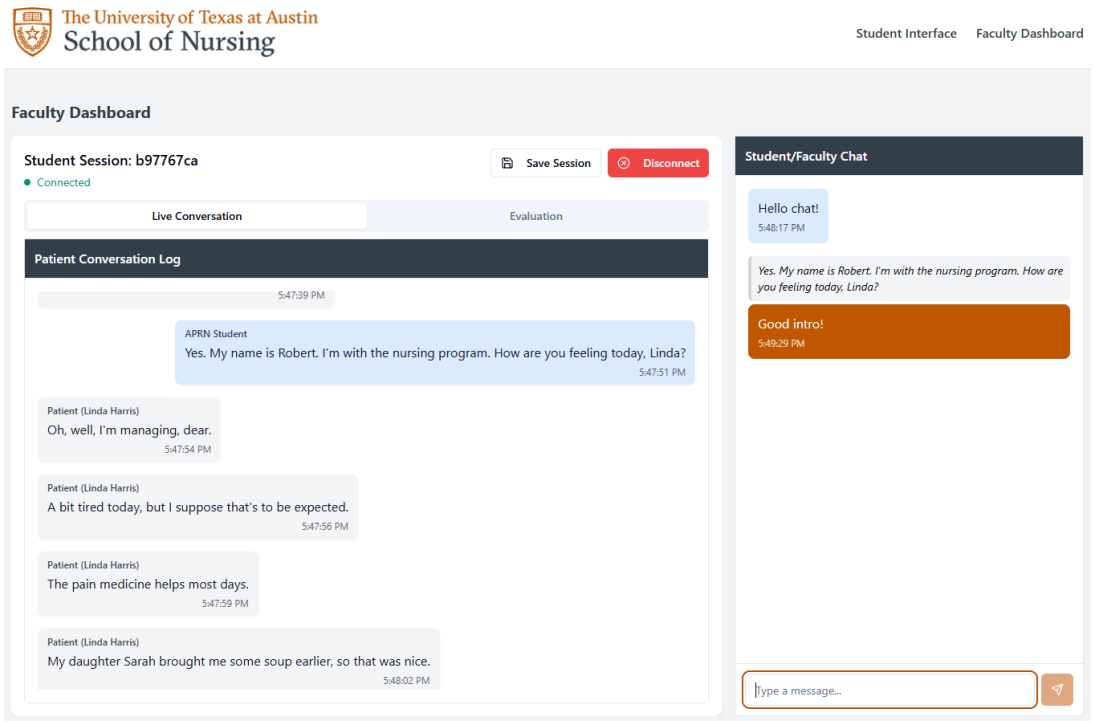
GUIDING THE AI: PATIENT PROFILES & STRATEGIES

- Faculty-defined Patient Profiles guide interactions (background, emotion, goals, sample phrases).
- AI uses strategies inspired by research (e.g., GuideLLM [1]) for:
  - **Goal Navigation:** Uses faculty goals/topics/milestones to guide AI responses towards learning objectives while following the student's lead.
  - **Context Management:** Includes recent conversation history in prompts for coherent dialogue.
  - **Empathetic Engagement:** Uses persona details (personality, emotion, style) to guide the AI's tone and maintain consistency.

[1] Duan et al. (2025) (https://arxiv.org/abs/2502.06494)

COMPLETED WORK (2-WEEK SPRINT)

- **Focus:** Functional Full-Stack Prototype
- **Key Achievements:**
  - Real-time server coordinating student-AI-faculty communication flow.
  - Faculty dashboard with live monitoring (audio + transcript).
  - Integration of emotional speech tools.
  - Scenario-driven personas & AI guidance logic.



ENGINEERING FOCUS & IMPACT

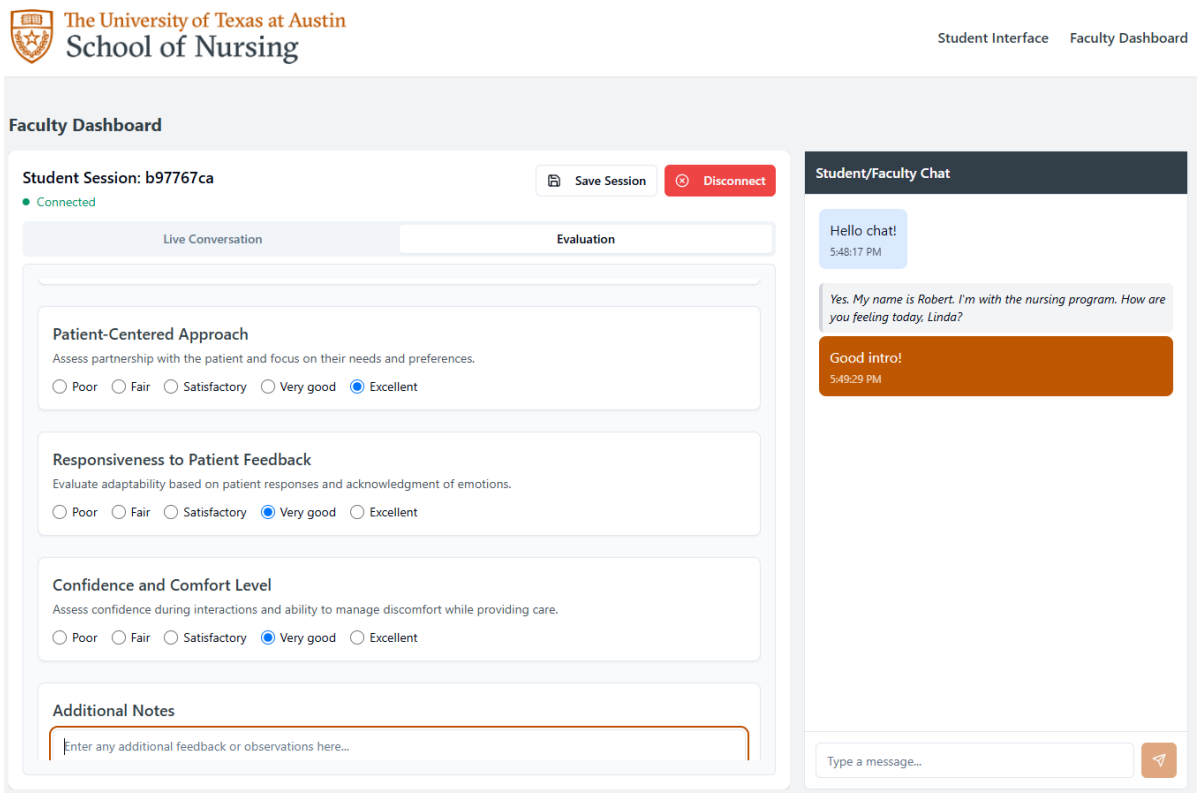
- Significant time on **backend engineering:**
  - Real-time WebSocket architecture.
  - API integrations.
  - Handling concurrent connections (student/faculty) - not native to LLMs.
- **Result:** Building this foundation left less time for refining emotional expression or conversational flow in the AI.

LESSONS LEARNED & FUTURE PIPELINE IDEA

- **Lesson:** Voice-to-Voice AI simplifies speech processing but offers limited fine-grained prompt control for emotional steering/expression.
- **Future Idea:** Modular AI Pipeline for Finer Control:
  - Speech-to-Text
  - Emotion/Sentiment Analysis (Student Voice)
  - Prompt Enrichment (Context + Emotion)
  - LLM Response Generation
  - Emotionally-Aware Speech Synthesis
- **Goal:** More natural, adaptive, and realistic AI responses.

FUTURE WORK & VISION

- **Extensible Design:**
  - Easily plug in newer/better LLMs (improves over time).
- **Planned Enhancements:**
  - Sophisticated emotion-aware modeling.
  - Expanded diverse scenario library (cultural, linguistic).
  - LMS integration (Canvas, Blackboard).
- **Long-Term:** Expressive video avatars (telehealth, non-verbal cues).



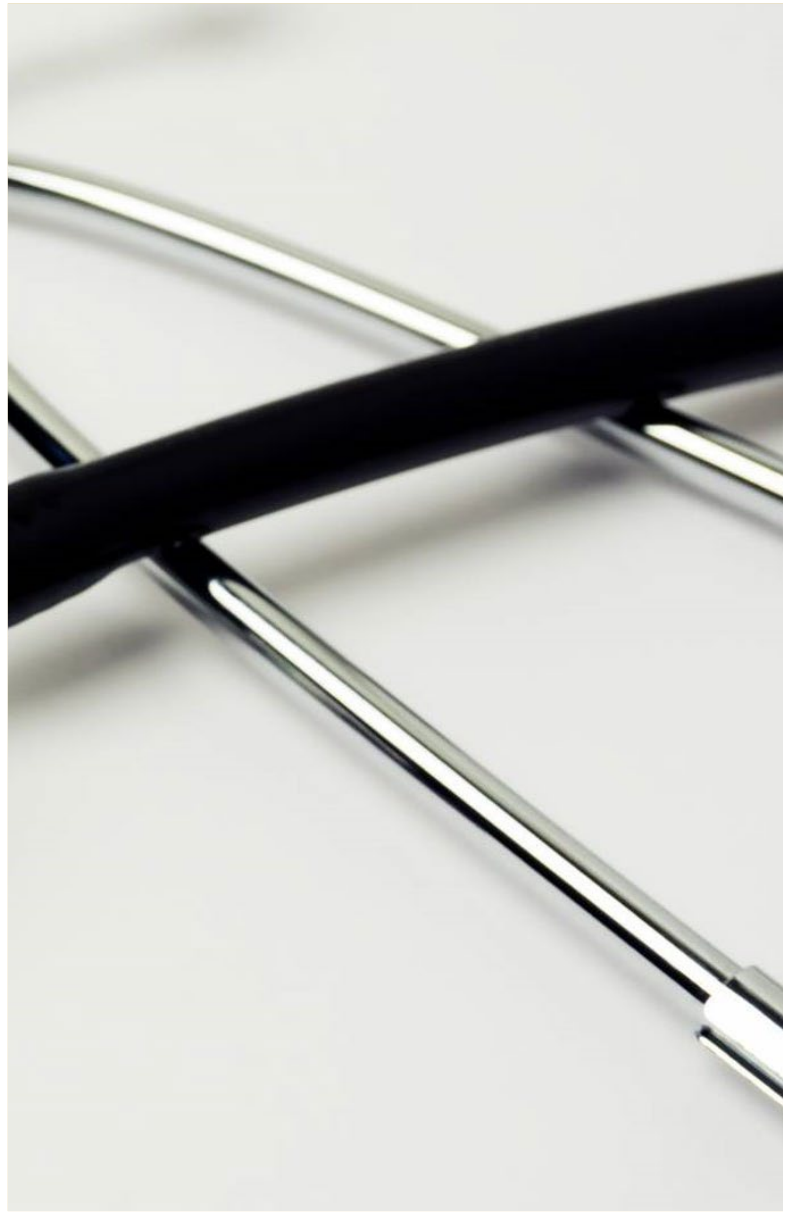
CORE GOAL & ETHICAL CONSIDERATIONS

- **Goal:** Practicing empathy, not just clinical checklists. Support realistic, flexible conversations.
- **Ethics:**
  - Uses synthetic patient data only (HIPAA not directly applicable now).
  - FERPA (student privacy) is crucial for evaluation/data storage.
  - Deployment requires robust privacy, transparency, compliance.

CLOSING SUMMARY

- **CareSpeak:** A tool designed with clinicians, for clinicians.
- Provides APRN students better practice for emotionally charged care.
- Focuses on realistic, empathetic voice interactions.





THANK YOU

Special thanks to Dr. Bo Xie, Dr. Cara Young and Dr. Ying Ding for all of the encouragement and support on this work!

Project page: https://github.com/rjleggieri/ut-nursing-ai

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